

Prasanna Sand

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Education

Santa Clara University - M.S. Computer Science and Engineering Santa Clara, CA	Sep 2023 – Jun 2025
Savitribai Phule Pune University - Master of Business Administration Pune, India	Jul 2021 – Aug 2023
MIT World Peace University - B.Tech. Computer Science and Engineering Pune, India	Jul 2018 – May 2021

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C++, Bash

AI / ML: RAG, OpenAI API, Gemini API, FAISS, Vector Databases, NLP, Computer Vision, TensorFlow, OpenCV

Frameworks & Libraries: React, Node.js, FastAPI, Flask, Express.js, Spring Boot, Angular, React Native

Infrastructure & DevOps: AWS (Lambda, EC2), Google Cloud, Firebase, Docker, Kubernetes, CI/CD (GitHub Actions)

Databases: MySQL, MongoDB, DuckDB, Redis, Firestore

Systems: REST APIs, Microservices, Distributed Systems, System Design, High-Concurrency Optimization

Experience

Software Engineer - Frugal Innovation Hub Santa Clara, CA	Dec 2024 – Present
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- Engineered production EdTech applications including a K-12 homework management platform and a bilingual (English/Spanish) science & math learning game and deployed to real students at underserved schools via non-profit partners.
- Architected modular backend services (presentation, domain, data layers) in an Agile development environment, with 85%+ unit test coverage enforced through rigorous code review standards.
- Automated CI/CD pipelines using GitHub Actions across multi-platform builds, cutting deployment time by ~40% and enabling reliable, rapid release cycles.
- Optimized API performance and database interactions, sustaining sub-200ms response times across 1K+ concurrent user sessions under real production load.
- Led 30+ peer code reviews and resolved 50+ pre-release issues, reducing post-deployment bugs by ~25% and authoring technical documentation to improve team-wide knowledge sharing.

Computer Vision Intern - NextLeap Aeronautics Bangalore, India	Jun 2020 – Aug 2020
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- Built a real-time UAV collision-prevention system using YOLOv3 in C++, achieving 92%+ mAP at 30 FPS; prototyped stereo-vision pipelines processing 60+ FPS for navigation at 20m/s.
- Improved power line detection precision by ~35% combining deep learning with classical vision techniques (Gabor filters, PCNF, Hough Transform).
- Explored multi-sensor fusion (vision + IMU + LiDAR), reducing localization drift by ~20% and improving environmental awareness in dynamic flight conditions.

Projects

Autonomous AI Agent System	Apr 2026
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- Engineered an autonomous AI agent using OpenAI API function calling, capable of decomposing high-level user goals into executable subtasks and orchestrating tool use across web search, code execution, and REST APIs with 95%+ task completion rate.
- Built a full-stack execution pipeline with FastAPI, Redis task queue, and MongoDB supporting 50+ concurrent agent runs, achieving end-to-end execution under 8 seconds; deployed via Docker and AWS Lambda with a React step-by-step execution dashboard.

AI-Powered Log Analysis & Anomaly Detection System	Apr 2026
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- Built an AI-powered observability system combining statistical anomaly detection (Z-score, sliding window) with OpenAI API-generated root cause analysis, surfacing real-time diagnostic insights from 8K+ log entries/minute.
- Designed full-stack pipeline from log ingestion (FastAPI) to LLM explanation layer and React monitoring dashboard, deployed on AWS with Docker achieving sub-150ms alert latency and 85%+ test coverage.

LLM-Powered Code Search & Recommendation Engine	Oct 2025
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- Engineered a semantic code retrieval system using vector embeddings and FAISS, reducing developer code search time and improving productivity across large codebases.
- Integrated LLMs to generate contextual code explanations and recommendations; exposed scalable REST APIs handling real-time queries with low latency.

AI-Powered Portfolio Assistant (RAG Chatbot)	Aug 2025
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- Built a production-ready intelligent chatbot using Retrieval-Augmented Generation (RAG) with FAISS similarity search to answer queries over structured and unstructured personal data.
- Deployed as a REST API with FastAPI, Docker, and AWS Lambda, delivering real-time source-attributed responses with scalable serverless inference.

All Bank – AI-Driven Inclusive Banking App	Nov 2024
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- Developed a full-stack financial management app using React Native and Firebase, offering goal-based savings, QR payments, and AI-powered financial insights for underbanked individuals and small businesses.
- Integrated Google Gemini AI and NLP to deliver real-time financial assistance, personalized analytics, and conversational support, enhancing accessibility and financial decision-making.

Certifications & Publications

Certifications: AWS Certified Solutions Architect - Associate | Certified Business Analytics | DeepLearning.AI TensorFlow Developer | Certified Google Cloud Engineer

Publications: [Brainwave Controlled Wheelchair](#) | [Applications of Computer Vision in Agriculture](#)